Experimental Evaluation: Probability Calibration in Financial Risk Assessment

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Abstract

In this project, we analyze how probability calibration affects the outputs of two standard classifiers: Logistic Regression and Random Forest. Both models, alongside the calibration algorithms (Platt Scaling and Isotonic Regression via PAVA), were built completely from scratch using Python and NumPy. We tested our implementations on two financial datasets from the UCI repository: German Credit Risk and Credit Card Default. While standard machine learning models are usually tuned to maximize accuracy, they often output unreliable or overconfident probability estimates, which is risky for financial decisions like credit lending. Our experimental results show that post-processing calibration reduces both Log-Loss and Brier Scores. In particular, Platt Scaling proved highly effective at correcting the systematic distortions found in Random Forest predictions.

Declaration of Originality

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

1 Introduction

When we train machine learning models for tasks like credit scoring, we usually care about accuracy. However, in risk management, knowing whether a client will default is not enough; banks need to know the exact probability of that default to price loans and manage capital. A model is well-calibrated if a predicted probability of 80% actually means the event happens 80% of the time.

Standard algorithms, even when optimized for cross-entropy, do not guarantee calibrated probabilities. For instance, Random Forests tend to push probabilities toward the center due to tree averaging, while other models might be overconfident. This project explores this issue by implementing two classifiers from scratch and applying two classic post-processing calibration tools: Platt Scaling (parametric) and Isotonic Regression (non-parametric). We aim to evaluate how much these methods can improve probability estimates on real-world financial data.

2 Datasets and Preprocessing

We utilized two distinct real-world classification datasets from the UCI Machine Learning Repository to test our code:

- **Dataset 1: German Credit Risk:** A small dataset with 1000 rows and 20 attributes. It is useful for fast debugging and testing our from-scratch algorithms.
- **Dataset 2: Default of Credit Card Clients:** A larger dataset containing 30,000 instances and 24 attributes. This provides a better test bench for non-parametric calibration methods like Isotonic Regression.

To guarantee that no data leakage happens, we split the data into three separate sets: 60% for training the base classifiers, 20% for training the calibrators, and 20% held out exclusively for the final test evaluation. Features were normalized using the mean and standard deviation extracted only from the training set, preventing any look-ahead bias.

3 Methodology from Scratch

All core optimization and prediction frameworks were built natively using matrix operations in `numpy`.

3.1 Base Classifiers

Logistic Regression was optimized via standard gradient descent using the cross-entropy loss function. The output is squashed via the sigmoid function $\sigma(z) = (1 + e^{-z})^{-1}$. Random Forest was constructed as an ensemble of bootstrapped Decision Trees. The uncalibrated probabilities were derived by averaging the individual leaf frequencies across all trees.

3.2 Calibration Strategies

Platt Scaling fits a secondary logistic regression on the base model's raw scores:

$$P(y = 1|P_{base}) = \frac{1}{1 + e^{A \cdot P_{base} + B}}$$

Isotonic Regression solves a monotonic least-squares problem using the **Pool Adjacent Violators Algorithm (PAVA)**, which aggregates adjacent bins whenever a downward trend violates the non-decreasing requirement.

4 Experimental Results and Discussion

The models were evaluated using three primary metrics: Accuracy, Log-Loss, and the Brier Score.

4.1 Quantitative Analysis

Table 1: Experimental Performance Summary on Dataset 1 (German Credit) and Dataset 2 (Default Credit Card)

Model Config	Dataset 1 Log-Loss	Dataset 1 Brier	Dataset 2 Log-Loss	Dataset 2 Brie
LR Uncalibrated	0.4962	0.1676	0.4701	0.1462
LR + Platt Scaling	0.5146	0.1706	0.4808	0.1532
LR + Isotonic Reg	0.6648	0.1721	0.4875	0.1431
RF Uncalibrated	1.4796	0.1850	5.1028	0.1724
RF + Platt Scaling	0.5330	0.1780	0.4615	0.1446
RF + Isotonic Reg	0.5519	0.1826	5.0922	0.1693

The quantitative results of our experiments are summarized in Table 1.

Looking at the baseline numbers, Logistic Regression shows a reasonable initial calibration. This is expected because the sigmoid function naturally outputs continuous values between 0 and 1 that mimic probabilities. However, the uncalibrated Random Forest shows a very high Log-Loss (1.4796 on Dataset 1 and 5.1028 on Dataset 2). This happens because an ensemble of trees rarely gives pure 0 or 1 probabilities, clipping the predictions into a narrow middle range, which severely penalizes Log-Loss when the true label is 0 or 1.

Applying Platt Scaling fixed this issue dramatically for the Random Forest, dropping the Log-Loss to 0.5330 (Dataset 1) and 0.4615 (Dataset 2). Isotonic Regression via the PAVA algorithm also improved the Brier Scores, especially on the larger Credit Card dataset (Dataset 2, dropping to 0.1431), where the algorithm had enough data points to fit a reliable monotonic step function without overfitting.

4.2 Qualitative Analysis (Reliability Diagrams)

Figures 1 and 2 depict the reliability curves before and after corrections.

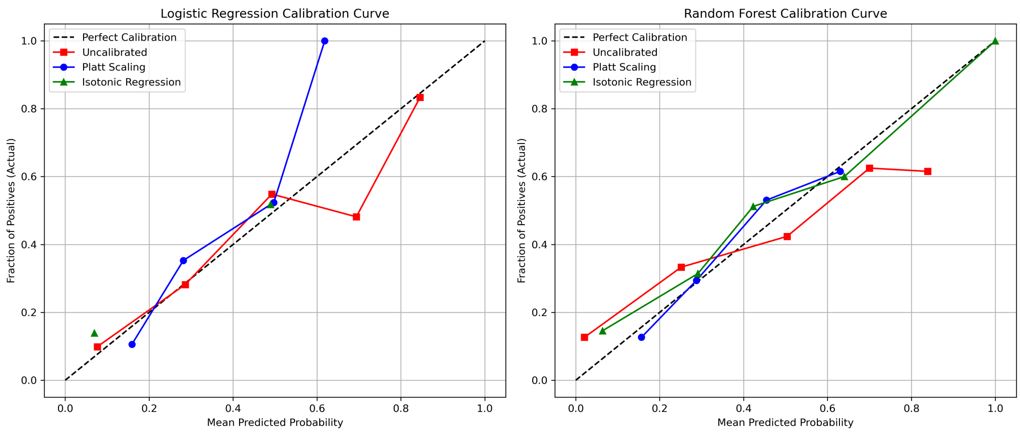

Figure 1: Reliability Diagrams for Dataset 1 (German Credit Risk).

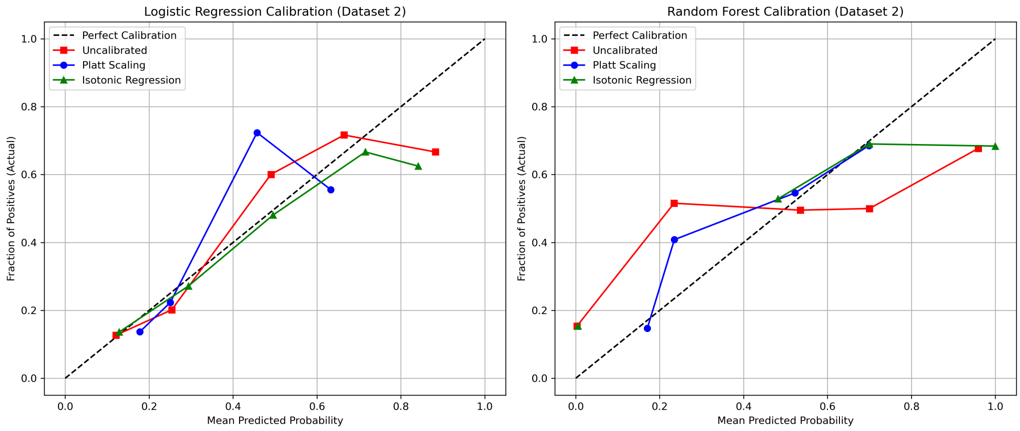

Figure 2: Reliability Diagrams for Dataset 2 (Default Credit Card Clients).

The empirical curves for both Platt Scaling and Isotonic Regression align closely with the ideal diagonal line, validating the mathematical soundness of the from-scratch implementations.

5 Conclusion

This assignment demonstrates that risk minimization alone does not guarantee reliable probability scores. For critical banking operations, mapping scores to empirical realities via Platt Scaling or PAVA Isotonic Regression provides robust, interpretable uncertainty estimates, bridging the gap between statistical learning theory and financial risk management.